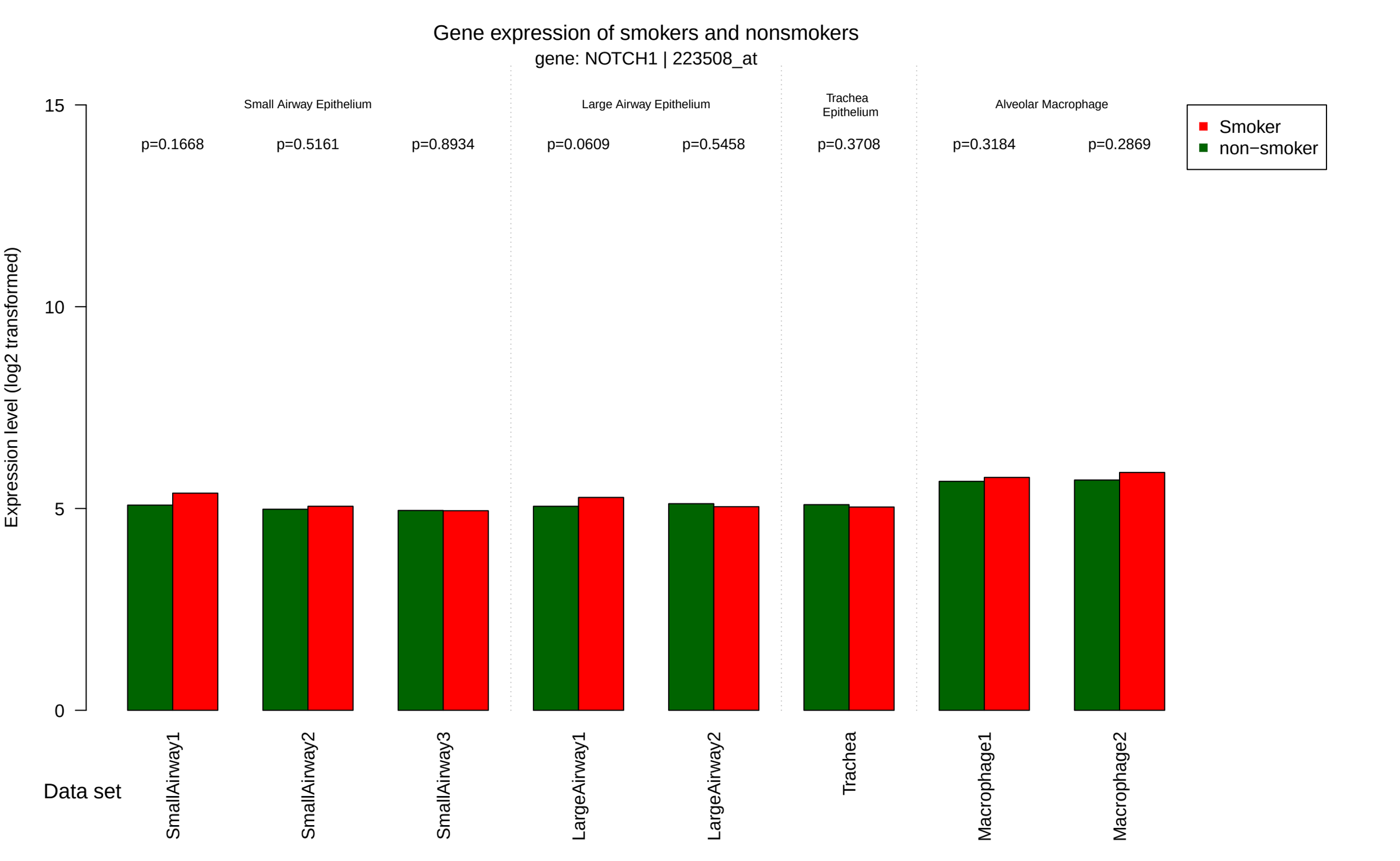
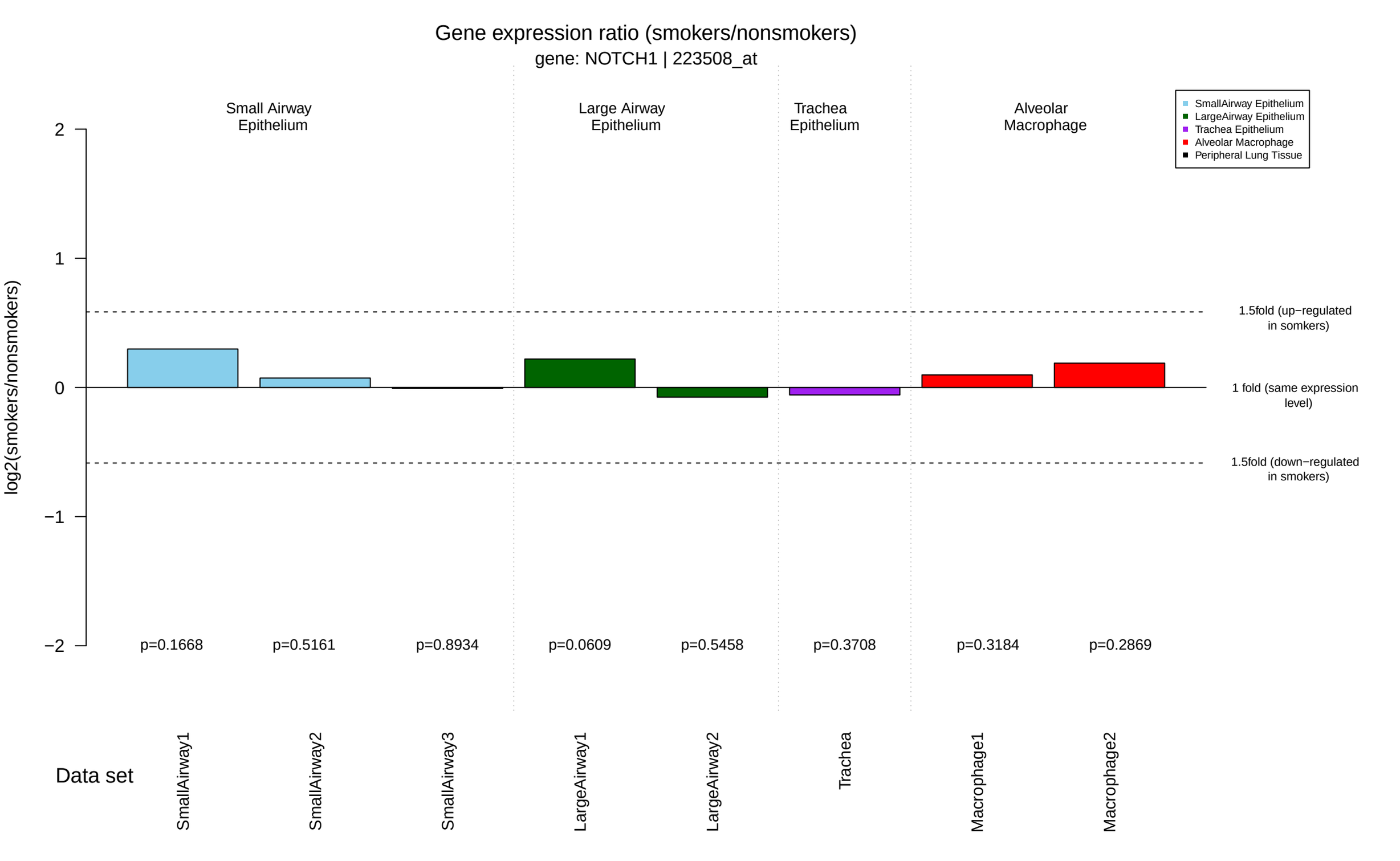
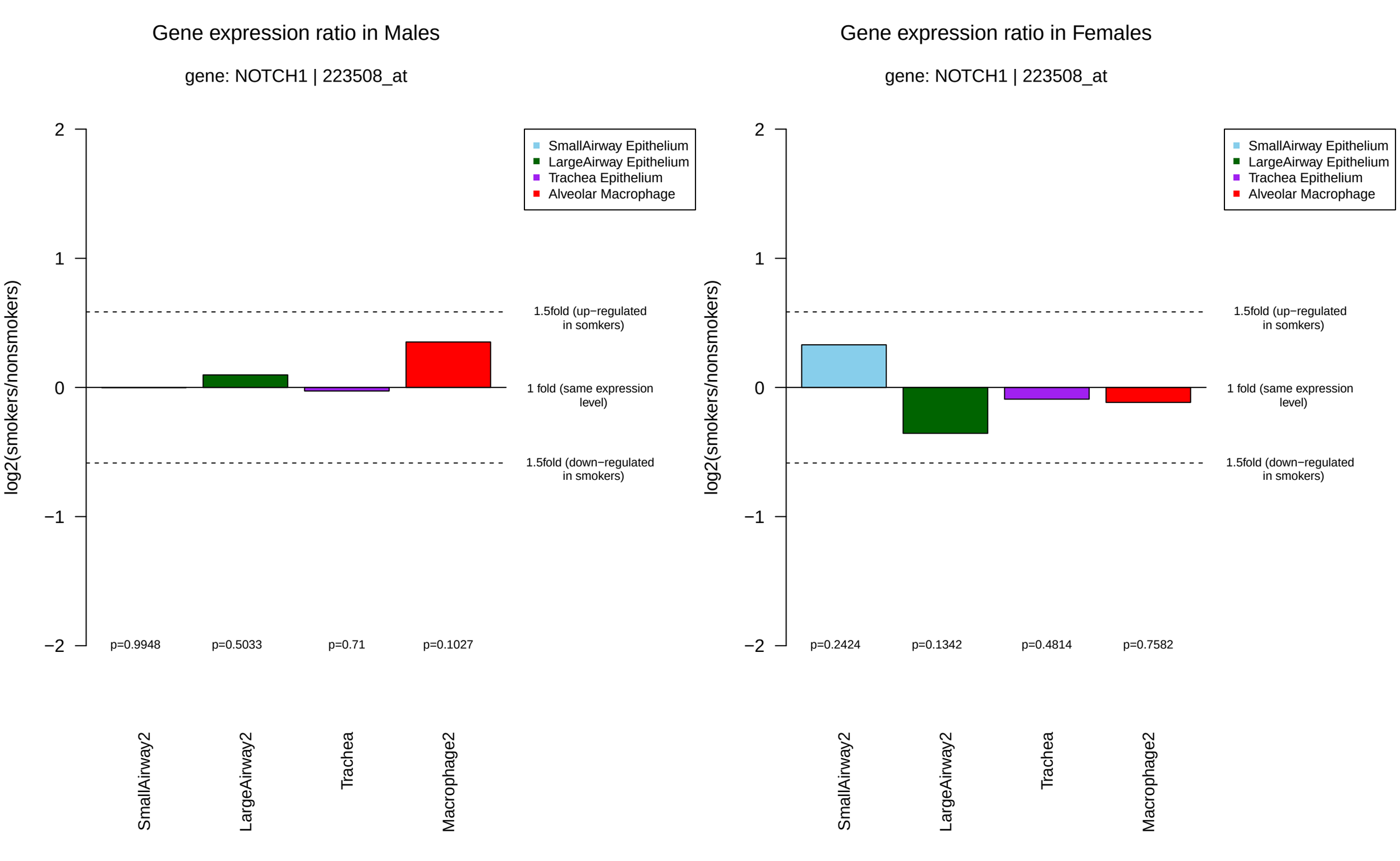
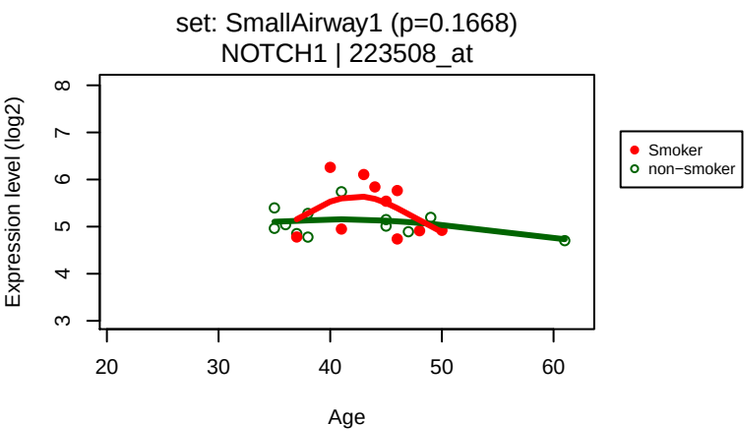
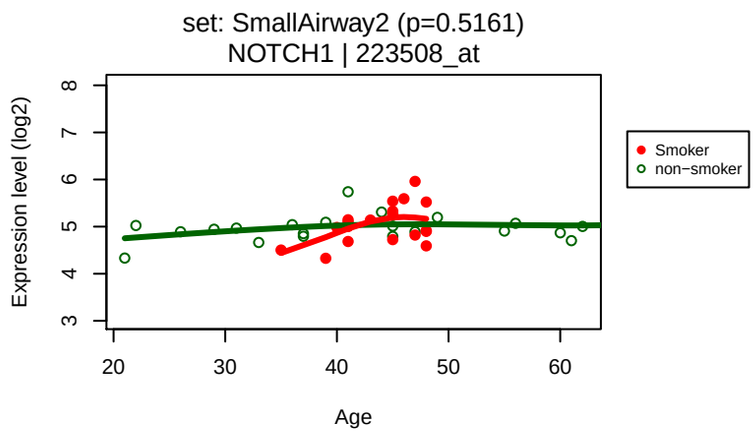
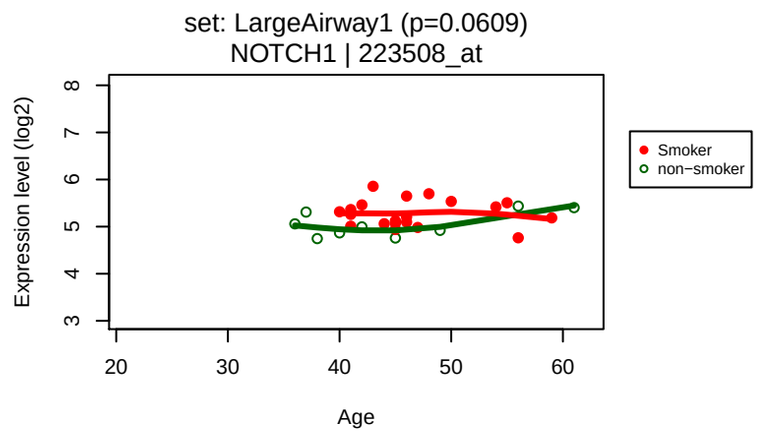

Gene expression of smokers and nonsmokers
gene: NOTCH1 | 223508_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1668
p=0.5161
p=0.8934
p=0.0609
p=0.5458
p=0.3708
p=0.3184
p=0.2869
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

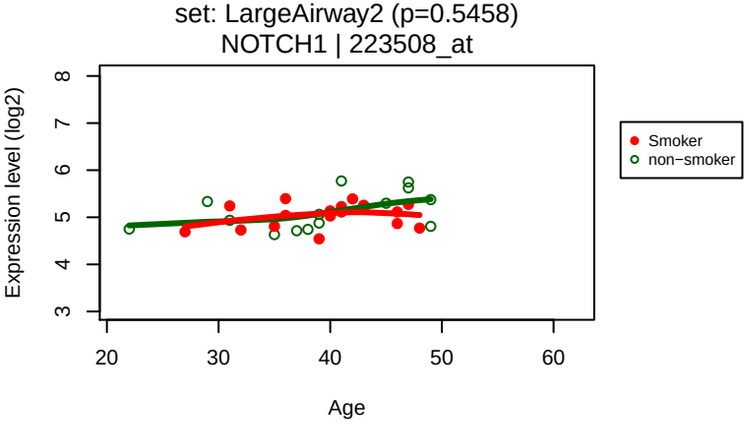

Gene expression ratio (smokers/nonsmokers)
gene: NOTCH1 | 223508_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1668
p=0.5161
p=0.8934
p=0.0609
p=0.5458
p=0.3708
p=0.3184
p=0.2869
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

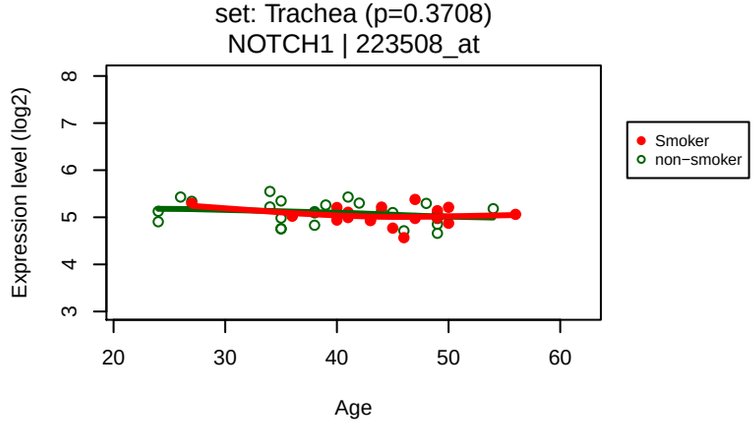

Gene expression ratio in Males
gene: NOTCH1 | 223508_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9948
p=0.5033
p=0.71
p=0.1027
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NOTCH1 | 223508_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2424
p=0.1342
p=0.4814
p=0.7582
SmallAirway2
LargeAirway2
Trachea
Macrophage2

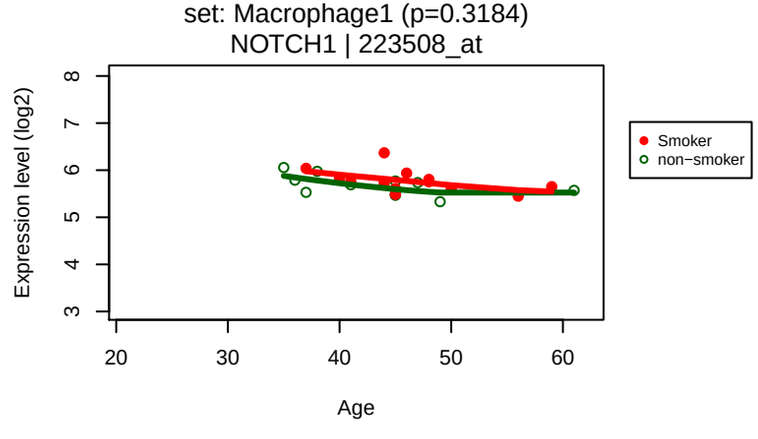

set: SmallAirway1 (p=0.1668)
NOTCH1 | 223508_at
Expression level (log2)
Age
Smoker
non−smoker

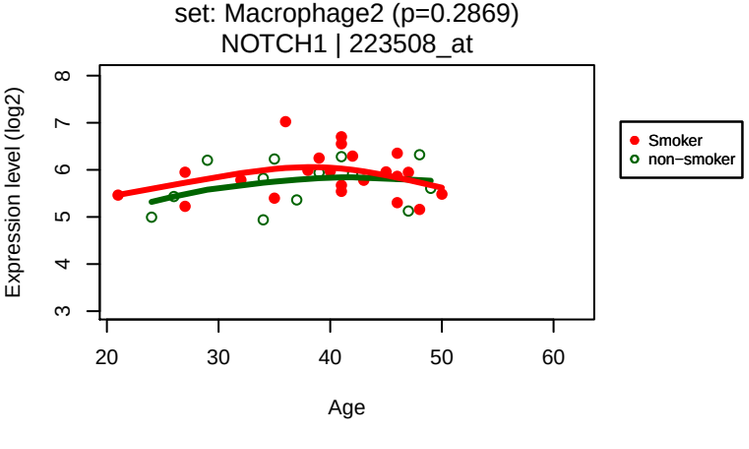

set: SmallAirway2 (p=0.5161)
NOTCH1 | 223508_at
Expression level (log2)
Age
Smoker
non−smoker

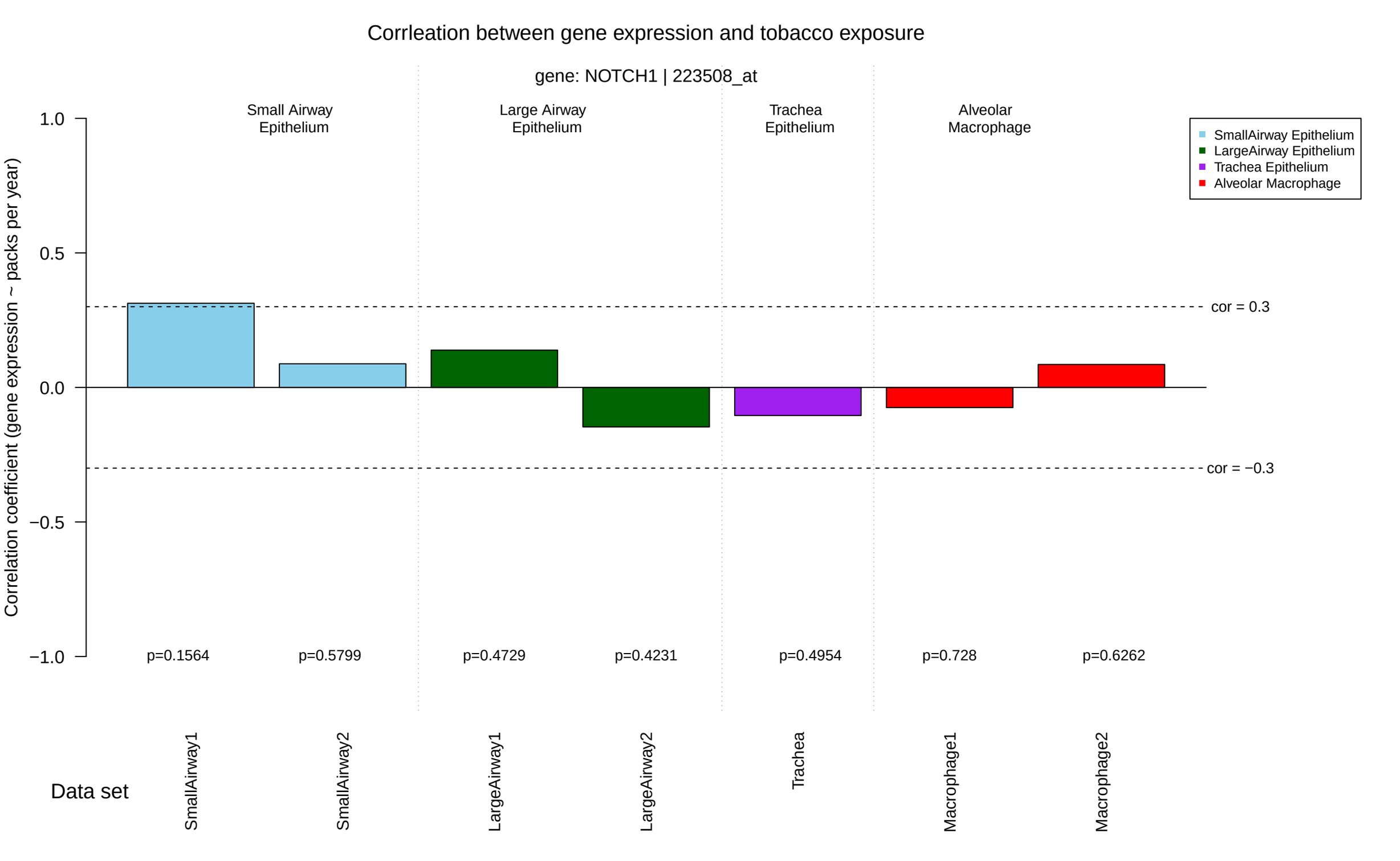

set: LargeAirway1 (p=0.0609)
NOTCH1 | 223508_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5458)
NOTCH1 | 223508_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3708)
NOTCH1 | 223508_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3184)
NOTCH1 | 223508_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2869)
NOTCH1 | 223508_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NOTCH1 | 223508_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1564
p=0.5799
p=0.4729
p=0.4231
p=0.4954
p=0.728
p=0.6262
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2